Customer Churn Prediction

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Executive Summary

A subscription provider can contact only a limited share of customers per retention cycle, so this project ranks customers by churn risk and explains the ranking instead of optimising accuracy. The analysis uses 1,200 customers and 20 raw fields, scikit-learn pipelines and a stratified 80/20 split (960 train / 240 test, seed 42). Results below come from the executed notebook; values read from its charts are marked *approx.* and values calculated from notebook outputs are marked *derived*.

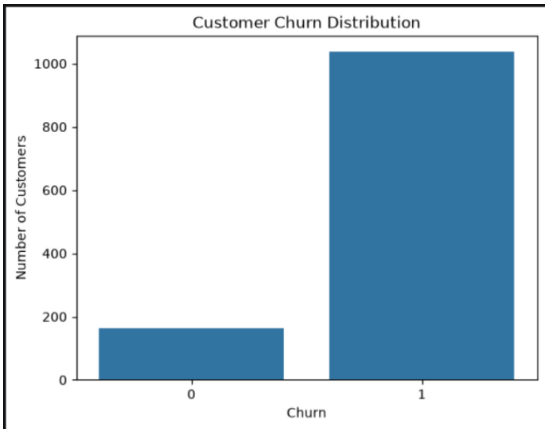
Headline	Finding
Churn base rate	86.3% churned, 13.7% retained. Unusually high; confirm label direction and population with the data owner.
Baseline models (test)	Logistic Regression ROC-AUC 0.9171, PR-AUC 0.9833, F1 0.9398. Random Forest (untuned) ROC-AUC 0.8878, PR-AUC 0.9764, F1 0.9258.
Tuned Random Forest	CV ROC-AUC 0.8736. Test at threshold 0.50: TN 26, FP 7, FN 17, TP 190 (derived: accuracy 0.900, precision 0.965, recall 0.918, F1 0.941). Average precision 0.9808.
Top-20% targeting	48 customers contacted, all churners; 23.2% of the 207 test churners captured vs about 41.4 (20%) expected at random: lift about 1.16x.
Cost-benefit (hypothetical)	Net +240 for model targeting vs about +174 for random contact (cost 5, value 100, 10% save rate).
Main risk pattern	Month-to-month contracts churn at about 91% vs about 79% on one- and two-year terms (approx.), and 88%-94% vs 76%-86% across payment methods.
Biggest gaps	Only 2 model families; no threshold sweep, SHAP/permutation output, customer-level explanations or robustness check (Section 8).

1.Data Audit and Preparation (Brief A)

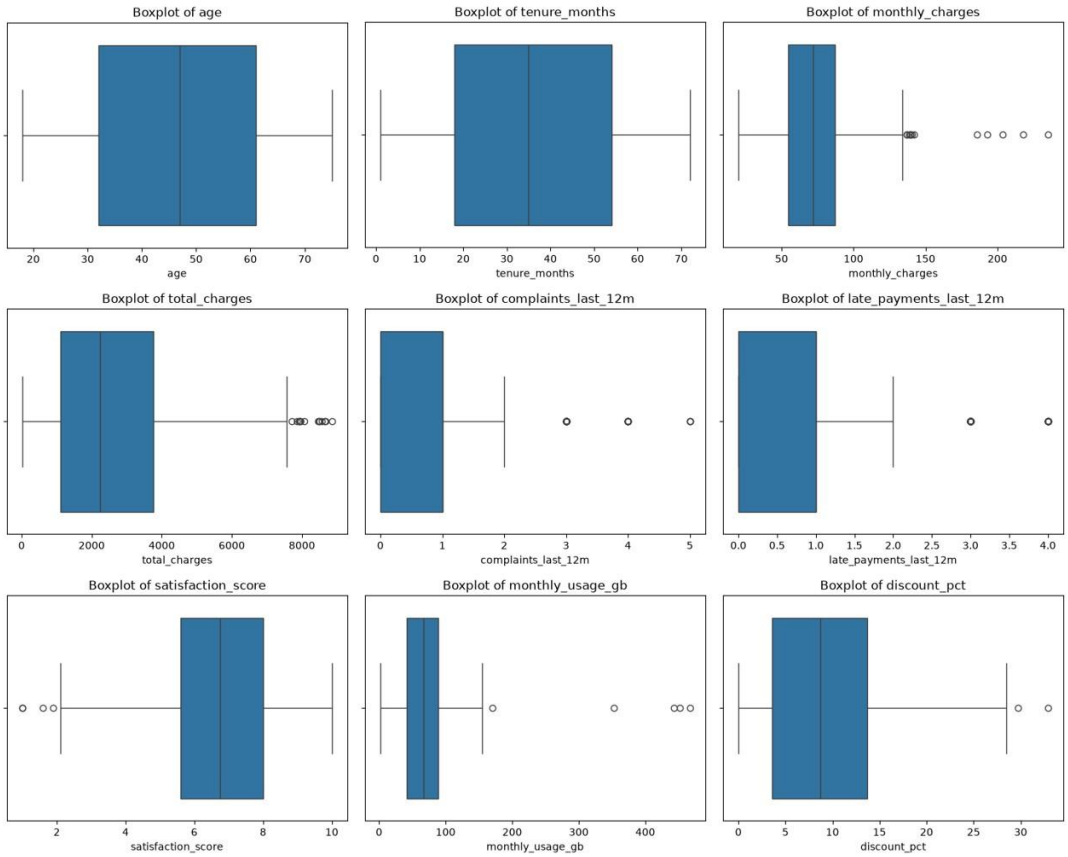
Checks	**Result**	**Desicion**
Shape / memory	1,200 rows x 20 columns, about 258 KB	-
Data types	5 float, 6 int, 9 text (incl. customer_id)	No corrections needed
Missing values	satisfaction_score 48 (4.0%), monthly_usage_gb 36 (3.0%), monthly_charges 30 (2.5%), total_charges 24 (2.0%); others complete	No rows dropped; median imputation inside the pipeline, fitted on training data only
Duplicates	0 duplicate rows, 0 duplicate customer_id	Nothing removed
Invalid values	Plausible ranges: age 18-75, tenure 1-72, satisfaction 1-10, counts 0-5 complaints / 0-4 late payments, discount 0 to about 23+	No rows removed
Outliers	Boxplots for 9 numeric fields (grid below). Monthly charges reach 235.01 vs 75th percentile about 87; satisfaction has low values 1-2.2	Kept as legitimate cases; no capping

Identifier	customer_id	Excluded from features

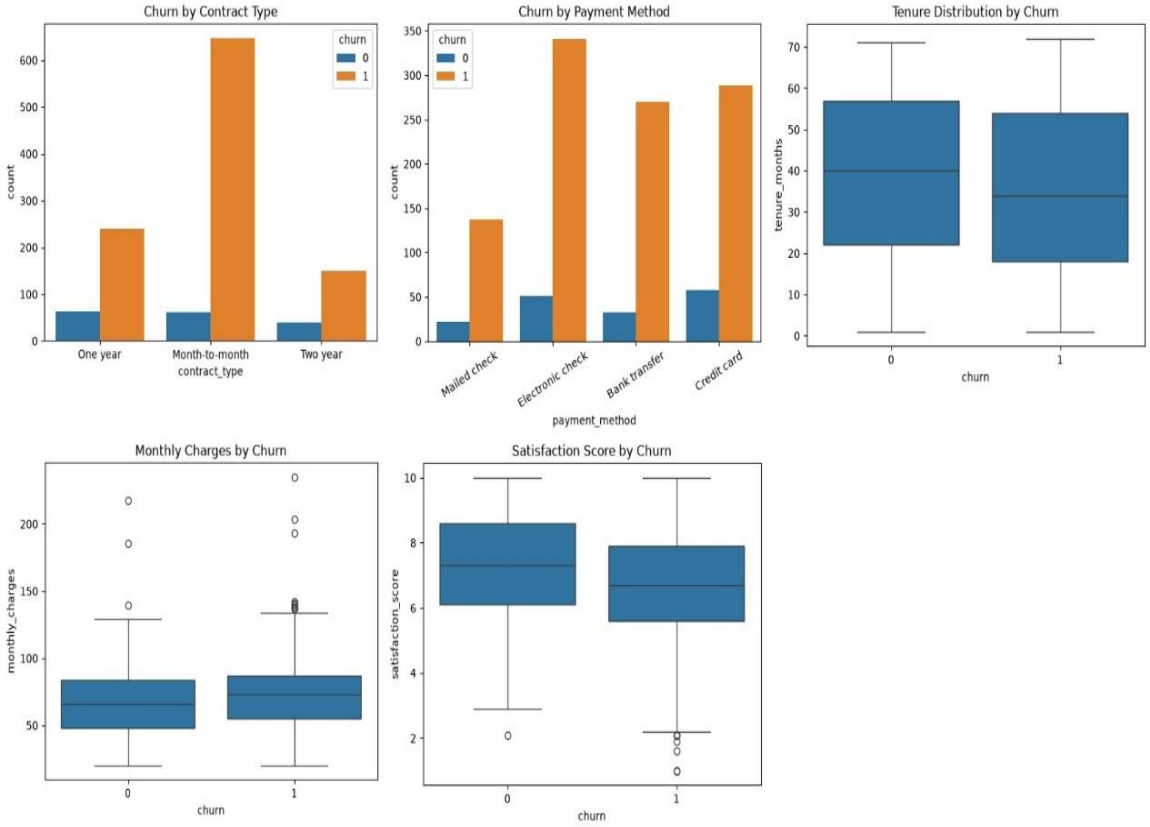
Target distribution. 86.3% churn vs 13.7% retained. A model that predicts churn for everyone would reach about 86% accuracy, so accuracy cannot separate a useful model from a trivial one. ROC-AUC, PR-AUC, precision/recall and top-20% lift are used instead. PR-AUC is also inflated by the high positive rate (baseline about 0.86).



Pipeline. Numeric: median imputer + StandardScaler. Categorical: most-frequent imputer + OneHotEncoder (unknown ignored). Combined in a ColumnTransformer and wrapped with each model in one Pipeline, so no test-set statistic enters preprocessing.



2.Exploratory Data Analysis (Brief B)

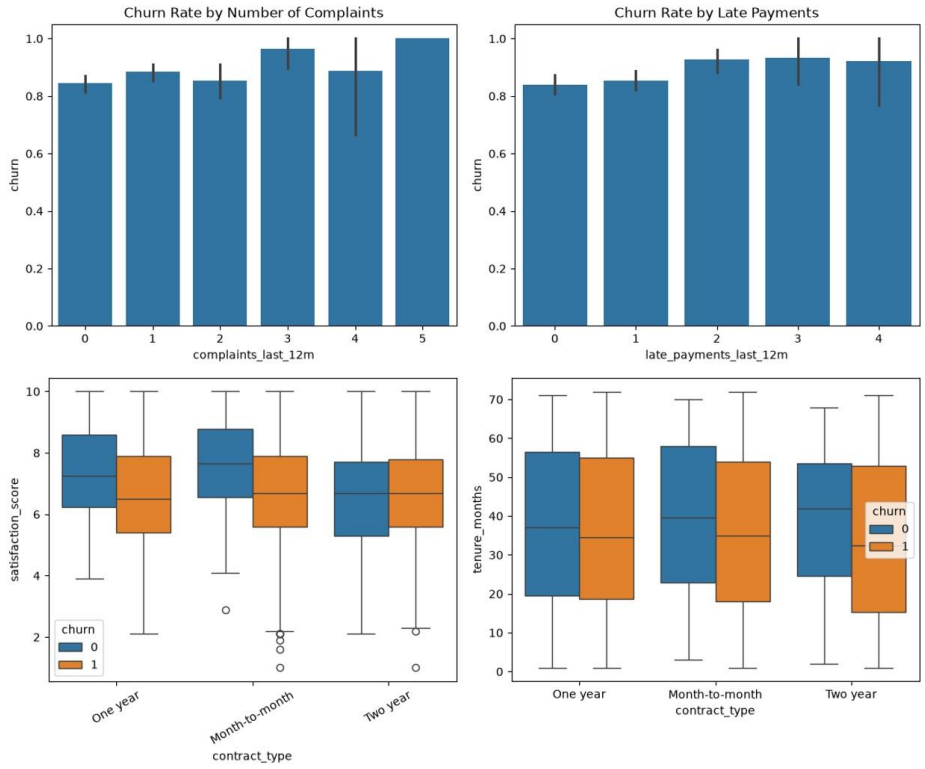


View	What the chart shows	Interpretation
Contract type	Churners about 648 vs 62 retained on month-to-month; about 240 vs 63 on one year; about 150 vs 40 on two year (approx.)	Month-to-month churn about 91%; longer terms about 79%. Strongest simple separator.
Payment method	Electronic check has the most customers (about 391); mailed check the fewest (about 159); churners dominate every group	Volume differs by method, but all groups are heavily churn; see interaction below for rate differences.
Tenure	Median about 40 months for retained vs about 34 for churners; similar spread	Churners are somewhat newer, but the boxes overlap heavily: a weak effect.
Monthly charges	Median about 66 retained vs about 73 churners; outliers above 130 in both groups	Slightly higher bills among churners; high-charge outliers occur in both classes.
Satisfaction	Median about 7.3 retained vs about 6.7 churners; many churners score 1-2.2	Lower satisfaction goes with churn; extreme low scores are almost all churners.

Nonlinear and threshold behaviour

Churn rate rises in steps rather than smoothly with complaints and late payments (next page), which supports the binary and count features engineered in Section 3 and favours tree models alongside the linear baseline.

Figure 3. Churn rate by complaints and late payments; satisfaction and tenure by contract type and churn (notebook plots).



View	What the chart shows	Interpretation
Complaints	Churn rate about 0.84 (0 complaints), 0.88 (1), 0.85 (2), 0.96 (3), 0.89 (4), 1.00 (5), with wide intervals above 2	Rises above 2 complaints, but the high-count groups are small, so treat as suggestive.
Late payments	About 0.84 (0), 0.85 (1), 0.93 (2), 0.93 (3), 0.92 (4)	Step up at 2 or more late payments: a threshold effect, not a straight line.
Contract x satisfaction	Retained customers rate satisfaction higher on month-to-month and one-year; on two-year medians are about equal (about 6.7)	On two-year terms satisfaction does not separate churners: the satisfaction effect depends on contract.
Contract x tenure	Churners have shorter tenure on every contract; gap largest on two-year (about 32 vs 42 median)	Short tenure matters most when customers are locked in a long contract.

Payment x contract churn rate (cross-tab output). Month-to-month is highest in every payment group (0.88-0.94); the top cells are bank transfer (0.936) and electronic check (0.930). One-year credit card (0.763) and two-year electronic check (0.770) are lowest. Differences within longer contracts are small and not consistently ordered.

Unusual observations. Monthly charges up to 235.01 and satisfaction of 1.0 are extreme but internally plausible (tenure and total charges are consistent), so they are treated as legitimate rather than errors. No formal outlier-sensitivity test was run.

3.Feature Engineering (Brief C)

Feature	Definition	Business / data rationale
charge_per_tenure	total_charges / tenure_months (0 tenure to NaN)	Average spend per month of tenure; separates price-heavy from long-standing customers
payment_risk	late_payments + complaints	One friction score for billing and service problems
complaint_intensity	complaints / (tenure + 1)	A complaint matters more for a new customer
has_late_payment	late_payments > 0	Any billing difficulty (the late-payment chart shows a step at 2+)
has_complaint	complaints > 0	Any service failure
low_satisfaction	satisfaction_score < 6	Flags clearly unhappy customers (low scores are mostly churners)
high_usage	monthly_usage_gb above 75th percentile	Heavy users may be more engaged or more price-sensitive

Leakage check. All seven features use only predictor columns, never churn, and none uses information from after the retention outcome. Two cautions: the high_usage cut-off was computed on the full dataset (mild leakage; compute it on training data only), and missing satisfaction/usage values silently become 0 in the two flags. No numeric leakage test (for example a single-feature AUC screen) was run.

4.Modeling (Brief D)

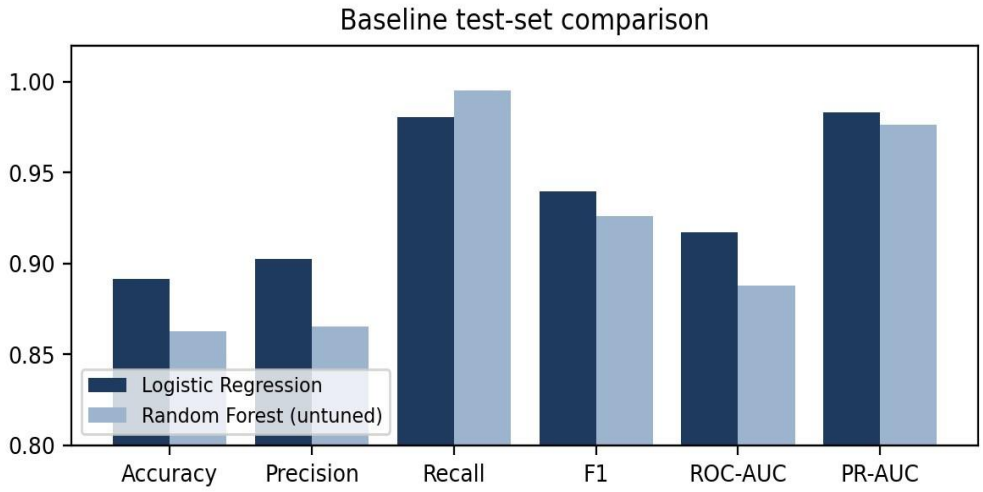


Figure 4. Baseline test-set comparison at threshold 0.50 (notebook output).

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression (baseline)	0.8917	0.9022	0.9807	0.9398	0.9171	0.9833
Random Forest, 300 trees (baseline)	0.8625	0.8655	0.9952	0.9258	0.8878	0.9764
Random Forest, tuned (derived from confusion matrix)	0.9000	0.9645	0.9179	0.9406	not printed	0.9808

- **Split:** stratified 80/20, seed 42; the test set was used for model comparison and final evaluation (it was not held back for a single final look, because baselines were compared on it).
- **Search:** RandomizedSearchCV, 5 candidates, 3-fold CV (15 fits), scoring ROC-AUC, seed 42. Grid: n_estimators {100,200,300}, max_depth {None,5,10,20}, min_samples_split {2,5,10}, min_samples_leaf {1,2,4}, class_weight {None, balanced}.
- **Best:** n_estimators 100, max_depth 10, min_samples_split 10, min_samples_leaf 2, class_weight balanced; CV ROC-AUC 0.8736. A fuller plan (20 iterations, 5-fold, average precision) was written but replaced by this lighter run.
- **Comparison:** tuning moved the Random Forest from predicting nearly everyone as churn (recall 0.995, precision 0.866) to a better balance (recall 0.918, precision 0.965). Its accuracy and F1 are now about equal to Logistic Regression, whose PR-AUC (0.9833) is slightly above the tuned forest's 0.9808.
- **Selection:** on documented metrics the two best candidates are practically tied, and Logistic Regression is simpler and more interpretable. The notebook saved the tuned Random Forest. Brief D asks for 3+ models; only 2 families were built, and Logistic Regression was never tuned.

5.Class Imbalance and Evaluation (Brief E)

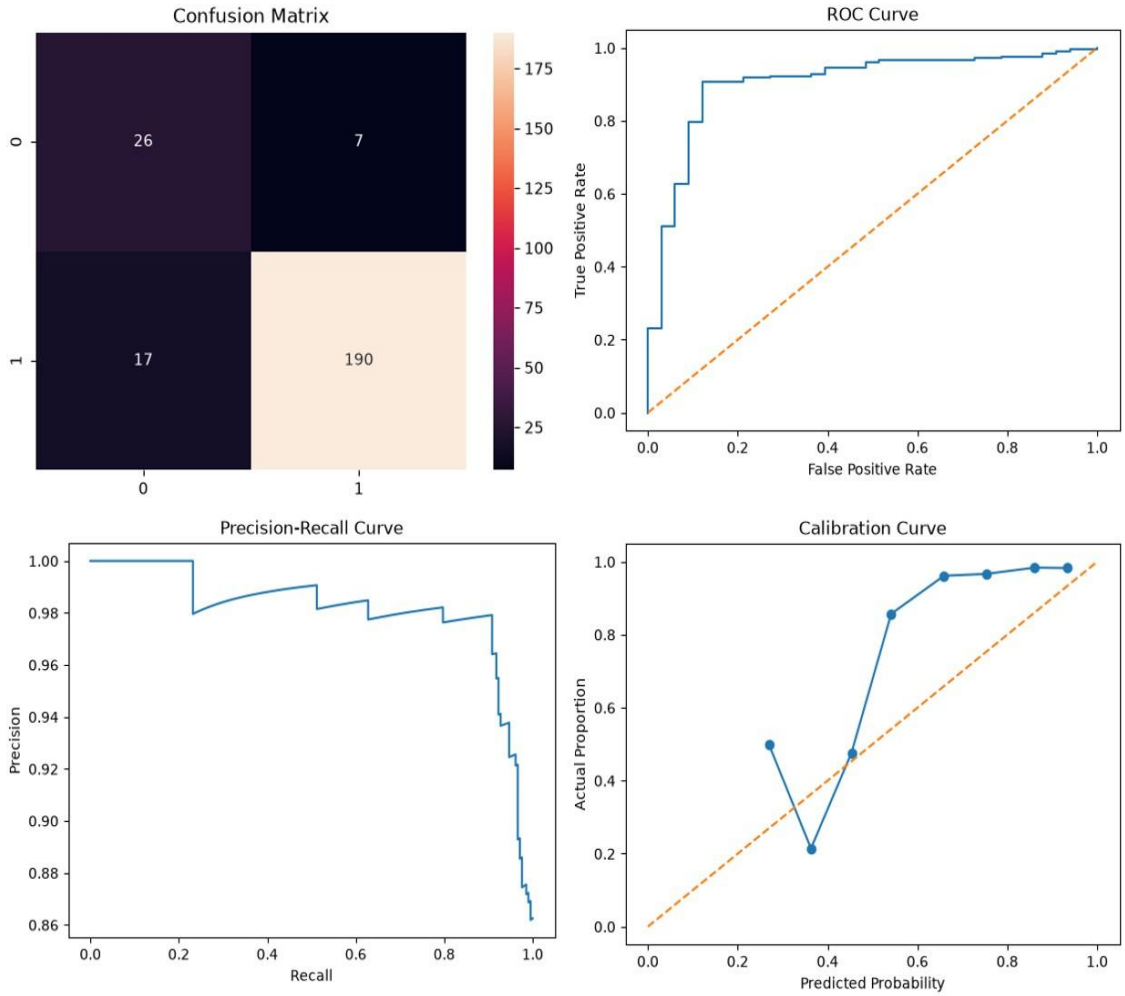


Figure 5. Tuned Random Forest on the 240-customer test set: confusion matrix (threshold 0.50), ROC curve, precision-recall curve, calibration curve (notebook plots).

Result	Value	Reading
Confusion matrix @0.50	TN 26, FP 7, FN 17, TP 190	Of 33 retained customers 26 are correct; of 207 churners 190 are found
Specificity (derived)	26/33 = 0.788	The model correctly clears about 79% of retained customers
Precision / recall / F1 (derived)	0.965 / 0.918 / 0.941	Few false alarms; 17 churners missed
Average precision	0.9808	High, but the no-skill level is about 0.86
ROC curve	TPR 0.23 at FPR 0; about 0.91 at FPR about 0.12	The top-ranked 48 customers are all churners; ranking is strong at the top

PR curve	Precision 1.0 up to recall about 0.23; at least 0.98 to recall about 0.8; falls to 0.86 at recall 1.0	Good ranking head; precision decays toward the base rate when most customers are contacted

Imbalance treatment and why accuracy is not enough

Churn is the majority class (86.3%); the minority is the retained group (13.7%). The notebook applied class_weight=balanced in the tuned forest and used no resampling, so nothing leaks through resampling. Logistic Regression was not re-run with class weights. Accuracy is insufficient because predicting churn for all customers already gives about 86%; the untuned forest's 0.8625 equals that rate.

Operating threshold

Using the hypothetical assumptions (contact cost 5, retained value 100, 10% save rate), contacting a customer pays off when P(churn) x 0.10 x 100 > 5, i.e. P(churn) > **0.50**. This derived break-even justifies 0.50 here, but only under those assumptions. Because most customers exceed it, the binding limit is the 20% contact capacity, so the practical rule is: contact the top 20% by score. The notebook defined thresholds 0.10-0.90 but did not compute precision/recall at each, so a threshold sweep table is not available.

Calibration

The calibration curve sits **above** the diagonal for predicted probabilities of about 0.54 to 0.93: observed churn there is about 0.86-0.98, so the class-weighted forest **under-states** churn probability in the upper range. In the low range (predicted 0.27-0.45) the points are non-monotonic (observed about 0.50, 0.21, 0.47), but those bins contain very few retained customers, so they are noisy. Conclusion: use the probabilities as a **ranking score**, and calibrate (isotonic or sigmoid, inside CV) before using them as true probabilities in financial calculations.

6.Explainability (Brief F)

A SHAP TreeExplainer was created on the tuned forest and permutation importance (average-precision scoring, 10 repeats, seed 42) was run on the test set, but **no SHAP values, importance table or customer-level explanations were printed**. Global importance and the three required individual explanations are therefore not available in the notebook. Until they are produced, the report's drivers rest on the EDA evidence in Section 2:
contract type, satisfaction, tenure, and complaint / late-payment counts.

Association vs causation. Importance describes what the model relies on to predict churn, not what causes it. Customers choose their contract and payment method, so a contract effect may reflect who selects it. Causal claims need a controlled pilot with a holdout group.

7.Retention Intervention Simulation (Brief G)

The 240 test customers were ranked by predicted churn probability and the top 20% (48) selected.

Quantity	Value
Customers contacted	48
Churners in test set (derived)	207
Capture rate (notebook)	0.2319 = 48/207
Contacted who churned (derived)	48 (all)
Random 20% expected (derived)	41.4
Lift (derived)	about 1.16x

The notebook simulated 100 random 20% draws but did not print their mean; the expected value 0.2 x 207 = 41.4 is used instead.

Cost-benefit (hypothetical: cost 5 per contact, value 100 per retained customer, 10% save rate)

	Model top 20%	Random 20% (expected)
Contact cost	48 x 5 = 240	48 x 5 = 240
Churners reached	48	41.4
Customers retained (10%)	4.8	4.14
Value retained	480	414
Net benefit	**+240**	**+174**

The model adds about +66 over random contact in this example. Only the contact cost was printed by the notebook; the rest is derived from its stated inputs. The result is highly sensitive to the 10% save rate: at 5% the model case only breaks even (value 240 vs cost 240) and random contact loses money.

8.Final Analysis (Brief H)

Churn-risk patterns supported by the data

- Month-to-month contracts churn at about 91% vs about 79% on one- and two-year terms (contract chart, approx.); the payment x contract table shows 0.88-0.94 vs 0.76-0.86.
- Highest-risk cells: month-to-month with bank transfer (0.936) or electronic check (0.930).
- Churners report lower satisfaction (median about 6.7 vs 7.3) and have somewhat shorter tenure (about 34 vs 40 months); on two-year contracts the satisfaction gap disappears while the tenure gap is largest.
- Churn rate steps up at 2+ late payments (0.84 to about 0.93) and at 3+ complaints (small groups, wide intervals).
- A ranking model finds all of the top 48 churners but gives only about 1.16x lift over random because 86% of customers churn.

Retention actions (evidence vs assumption)

Action	Evidence from data	Business assumption (untested)
1. Offer one- or two-year renewal incentives to month-to-month customers	About 12 points lower churn on longer terms (91% vs 79%, approx.)	Customers accept commitment and the lower churn is caused by the contract
2. Prioritise month-to-month bank transfer / e-check customers	Highest cells 0.936 and 0.930	These customers respond at the assumed 10% save rate
3. Add billing support (autopay reminders, flexible due dates) for customers with 2+ late payments	Churn about 0.93 at 2+ late payments vs 0.84-0.85 at 0-1	Billing friction causes churn and support reduces it
4. Service-recovery call for customers with 3+ complaints or satisfaction below 6	Churn 0.96+ at 3+ complaints (small n); low scores mostly churners	Fast resolution lowers churn
5. Run a controlled pilot (contacted vs holdout) before scaling	Lift is modest; save rate unknown	Pilot replaces the 10% assumption

Compliance with the brief

Requirement	Status
A Data audit, pipeline, customer_id as identifier	Done
B 6+ visualisations with interpretation; interactions; unusual cases	Done (7 single-factor plots, 3 interaction views, 9 boxplots)
C 5+ features with rationale; leakage argument	Done; leakage argued, no numeric test; high_usage cut-off uses full data
D 3+ models, CV tuning, baseline vs tuned	Partial: 2 models; Logistic Regression untuned; light 3-fold search
E Metrics, confusion matrix, threshold from business assumption	Done for metrics and matrix; threshold sweep not run; break-even 0.50 derived here
F Global importance and 3 customer explanations	Not done (no output)
G Top-20% vs random; cost-benefit	Done (random mean and cost-benefit totals derived)
Adv. Precision/recall across thresholds	Curve drawn; no threshold table
Adv. Calibration discussion	Done in this report (curve from notebook)
Adv. Robustness check (fold variation or outlier test)	Not done
Adv. Seeds, versions, steps	Seed 42, Python 3.13.5, pandas 3.x documented; sklearn/shap versions not recorded
Deliverables 01-03 (notebook, cleaned CSV, model)	Notebook saves ../data/02_cleaned_dataset.csv and ../models/03_model_artifact.joblib
Deliverable 05 README.md	Not provided

Execution steps. Load data, audit, EDA, feature engineering, stratified split (seed 42), build pipelines, fit baselines, tune the Random Forest, evaluate, run the top-20% simulation, then save the model and cleaned dataset.